

Abstract Algebra - Homework 2

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Problem 1

Let $|G| = p^\alpha m$, where p is a prime, $\alpha \geq 1$ and $p \nmid m$. By definition of a p -Sylow subgroup, we have $|P| = p^\alpha$. Since $N \triangleleft G$, Lagrange's theorem implies $|N| \mid |G|$. Thus, we can write:

$$|N| = p^\beta d,$$

with $\beta \in [1, \alpha]$ and $d \mid m$, where $p \nmid d$. Consider the subgroup $N \cap P$. Since $N \cap P \leq P$, again applying Lagrange's theorem gives $|N \cap P| \mid |P| = p^\alpha$. Therefore, we must have:

$$|N \cap P| = p^\gamma,$$

where $\gamma \leq \beta$, hence $N \cap P$ must be a p -subgroup of N .

Consider the subgroup NP which contains N and P . Since $N \triangleleft G$ and $P \leq G$ we may apply the following formula,

$$|NP| = \frac{|N||P|}{|N \cap P|} = \frac{p^{\alpha+\beta}d}{p^\gamma} = p^{\alpha+\beta-\gamma}d$$

Since P is a p -Sylow subgroup of G it must also be a p -Sylow subgroup in any subgroup of G containing it. Therefore, its index in NP must be relatively prime to p .

$$[NP : P] = \frac{|NP|}{|P|} = \frac{p^{\alpha+\beta-\gamma}d}{p^\alpha} = p^{\beta-\gamma}d \implies p^{\beta-\gamma} = 1 \implies \beta = \gamma.$$

This shows $|N \cap P| = p^\beta$, which means it is a p -Sylow subgroup of N .

The quotient group G/N inherits a finite order, $|G/N| = \frac{|G|}{|N|} = \frac{p^\alpha m}{p^\beta d} = p^{\alpha-\beta} \frac{m}{d}$. A p -Sylow subgroup must therefore be of order $p^{\alpha-\beta}$. Since we

concluded $\gamma = \beta$, $|NP| = p^\alpha d$. Moreover $|N| = p^\beta d$. This allows us to construct a candidate subgroup with the correct order, NP/N

$$|NP/N| = \frac{|NP|}{|N|} = \frac{p^\alpha d}{p^\beta d} = p^{\alpha-\beta}$$

Finally, since NP is a subgroup of G containing N , the quotient NP/N is a subgroup of G/N . Therefore, NP/N is a subgroup of order $p^{\alpha-\beta}$ in G/N , and hence is a p -Sylow subgroup of G/N .

Problem 2

- (1) Naturally, since both P and Q are Sylow p -subgroups of the finite group G , the well-known Sylow theorem assures us P and Q are conjugate.
- (2) ($\subseteq$) Let $g \in N_G(P)$, so by definition $gPg^{-1} = P$. Since f is a homomorphism, we have:

$$f(g)f(P)f(g)^{-1} = f(gPg^{-1}) = f(P),$$

which implies $f(g) \in N_H(f(P))$. Therefore, $f(N_G(P)) \subseteq N_H(f(P))$.

($\supseteq$) Now let $h \in N_H(f(P))$. Since f is surjective, there exists $g \in G$ such that $f(g) = h$. Consider the conjugate subgroup $gPg^{-1} \leq G$. Then:

$$f(gPg^{-1}) = f(g)f(P)f(g)^{-1} = hf(P)h^{-1} = f(P),$$

so $f(gPg^{-1}) = f(P)$. By Problem 2.1, it follows that gPg^{-1} and P are conjugate in G ; that is, there exists $x \in G$ such that $xgPg^{-1}x^{-1} = P$. Define $y = xg$, then:

$$yPy^{-1} = xgPg^{-1}x^{-1} = P \Rightarrow y \in N_G(P),$$

and since $f(y) = f(xg) = f(x)f(g)$, we get:

$$f(g) = f(x)^{-1}f(y) \in f(N_G(P)).$$

Thus $h = f(g) \in f(N_G(P))$, so $N_H(f(P)) \subseteq f(N_G(P))$.

- (3) Assume that p does not divide the order of H . Let P be a p -Sylow subgroup of G .

Since $|f(P)|$ divides $|H|$ and $p \nmid |H|$, it follows that $|f(P)| = 1$, i.e.,

$$f(P) \subseteq \ker(f).$$

From Problem 2.2, we know that $f(N_G(P)) = N_H(f(P))$. Since $f(P)$ is trivial, its normaliser is all of H , so:

$$N_H(f(P)) = H \Rightarrow f(N_G(P)) = H.$$

Problem 3

Problem 4

Problem 5

Problem 6